In [31]:
```
import pandas as pd
```

In [23]:
```
dataset=pd.read_csv("insurance.csv")
```

In [24]:
```
dataset
```

Out[24]:

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520
...	...	...	...	...	...	...	...
1333	50	male	30.970	3	no	northwest	10600.54830
1334	18	female	31.920	0	no	northeast	2205.98080
1335	18	female	36.850	0	no	southeast	1629.83350
1336	21	female	25.800	0	no	southwest	2007.94500
1337	61	female	29.070	0	yes	northwest	29141.36030

1338 rows × 7 columns

In [33]:
```
df = pd.read_csv("insurance.csv.xls")
```

In [34]:
```
print(df.head())
```
```
   age     sex     bmi  children smoker     region      charges
0   19  female  27.900         0    yes  southwest  16884.92400
1   18    male  33.770         1     no  southeast   1725.55230
2   28    male  33.000         3     no  southeast   4449.46200
3   33    male  22.705         0     no  northwest  21984.47061
4   32    male  28.880         0     no  northwest   3866.85520
```

In [35]:
```
print(df.columns)
```
```
Index(['age', 'sex', 'bmi', 'children', 'smoker', 'region', 'charges'], dtype='str')
```

In [36]:
```
print(df.shape)
```
```
(1338, 7)
```

In [38]:
```
df = pd.read_csv("insurance.csv.xls")
```

In [39]:
```
print(df.head())
print(df.shape)
print(df.columns)
print(df.isnull().sum())
```
```
   age     sex     bmi  children smoker     region      charges
0   19  female  27.900         0    yes  southwest  16884.92400
1   18    male  33.770         1     no  southeast   1725.55230
2   28    male  33.000         3     no  southeast   4449.46200
3   33    male  22.705         0     no  northwest  21984.47061
4   32    male  28.880         0     no  northwest   3866.85520
(1338, 7)
Index(['age', 'sex', 'bmi', 'children', 'smoker', 'region', 'charges'], dtype='str')
age         0
sex         0
bmi         0
children    0
smoker      0
region      0
charges     0
dtype: int64
```

In [40]:
```
X = df.drop("charges", axis=1)
```

In [41]:
```
y = df["charges"]
```

In [42]:
```
print(X.head())
print(y.head())
```
```
   age     sex     bmi  children smoker     region
0   19  female  27.900         0    yes  southwest
1   18    male  33.770         1     no  southeast
2   28    male  33.000         3     no  southeast
3   33    male  22.705         0     no  northwest
4   32    male  28.880         0     no  northwest
0    16884.92400
1     1725.55230
2     4449.46200
3    21984.47061
4     3866.85520
Name: charges, dtype: float64
```

In [43]:
```
numerical_cols = ["age", "bmi", "children"]

categorical_cols = ["sex", "smoker", "region"]
```

In [44]:
```
numeric_preprocessor = StandardScaler()

categorical_preprocessor = OneHotEncoder(
    handle_unknown="ignore"
)

preprocessor = ColumnTransformer(
    transformers=[
        ("num", numeric_preprocessor, numerical_cols),
        ("cat", categorical_preprocessor, categorical_cols)
    ]
)
```

In [45]:
```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

print("Training data:", X_train.shape)
print("Testing data:", X_test.shape)
```
```
Training data: (1070, 6)
Testing data: (268, 6)
```

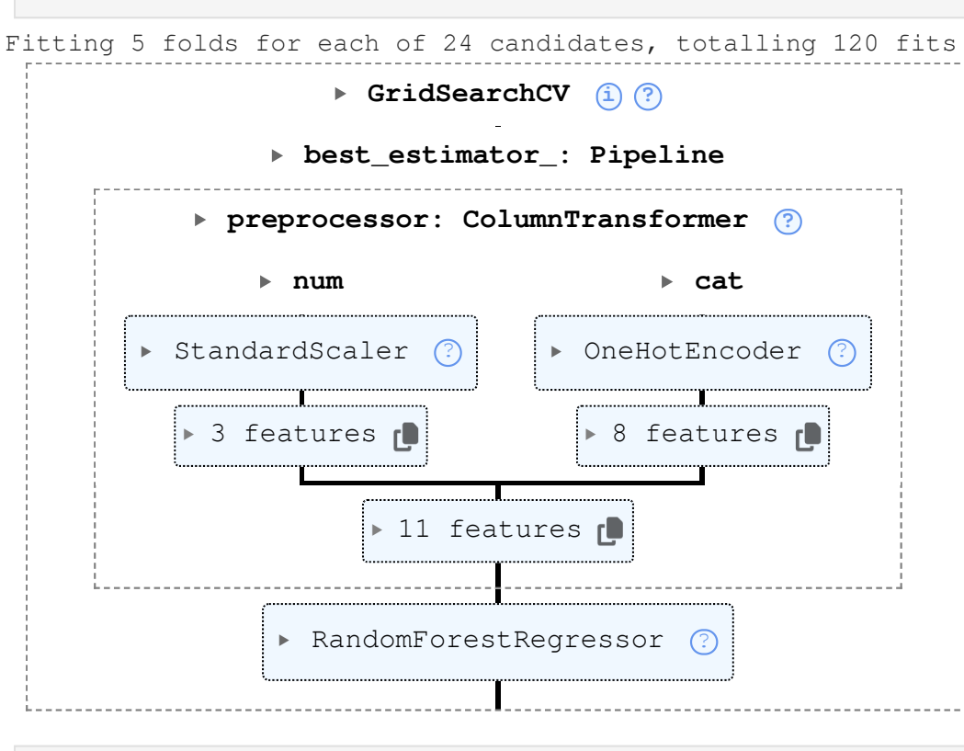
In [46]:
```
rf_pipeline = Pipeline([
    ("preprocessor", preprocessor),
    ("model", RandomForestRegressor(random_state=42))
])
```

In [47]:
```
rf_params = {
    "model__n_estimators": [100, 200],
    "model__max_depth": [None, 10, 20],
    "model__min_samples_split": [2, 5],
    "model__min_samples_leaf": [1, 2]
}
```

In [48]:
```
rf_grid = GridSearchCV(
    estimator=rf_pipeline,
    param_grid=rf_params,
    cv=5,
    scoring="r2",
    n_jobs=-1,
    verbose=1
)
```

In [49]:
```
rf_grid.fit(X_train, y_train)
```
```
Fitting 5 folds for each of 24 candidates, totalling 120 fits
```

Out[49]:



In [50]:
```
print("Best RF Parameters:", rf_grid.best_params_)
print("Best RF CV Score:", rf_grid.best_score_)
```
```
Best RF Parameters: {'model__max_depth': 10, 'model__min_samples_leaf': 2, 'model__min_samples_split': 5, 'model__n_estimators': 200}
Best RF CV Score: 0.8386156755486546
```

In [51]:
```
rf_pred = rf_grid.predict(X_test)

print(rf_pred[:10])
```
```
[10149.67221128  5143.92551025 27851.89866769 11092.3661993
 34582.9906328   7415.14216856  2054.67285512 14789.54006581
  6488.96252037 11305.82411486]
```

In [52]:
```
print("MAE:", mean_absolute_error(y_test, rf_pred))

print("RMSE:", np.sqrt(mean_squared_error(y_test, rf_pred)))

print("R2 Score:", r2_score(y_test, rf_pred))
```
```
MAE: 2433.3858829339824
RMSE: 4419.815912644427
R2 Score: 0.8741711283953388
```

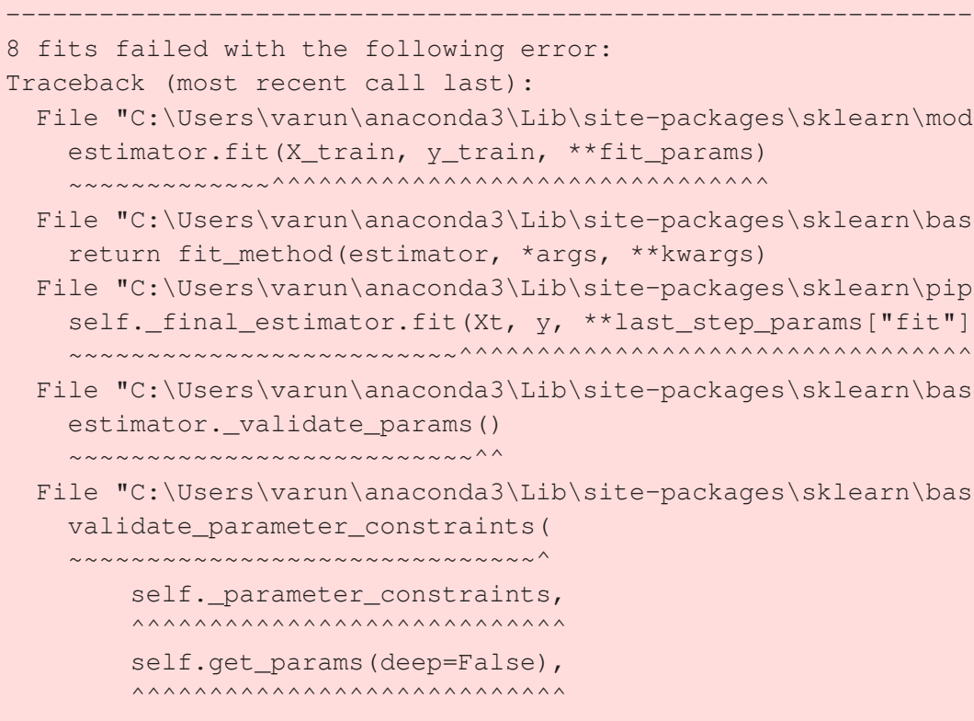
In [53]:
```
dt_pipeline = Pipeline([
    ("preprocessor", preprocessor),
    ("model", DecisionTreeRegressor(random_state=42))
])
```

In [54]:
```
dt_params = {
    "model__max_depth": [None, 5, 10, 20],
    "model__min_samples_split": [2, 5, 10],
    "model__min_samples_leaf": [1, 2, 4],
    "model__criterion": ["squared_error", "friedman_mse"]
}
```

In [55]:
```
dt_grid = GridSearchCV(
    estimator=dt_pipeline,
    param_grid=dt_params,
    cv=5,
    scoring="r2",
    n_jobs=-1,
    verbose=1
)
```

In [56]:
```
dt_grid.fit(X_train, y_train)
```
```
Fitting 5 folds for each of 72 candidates, totalling 360 fits
```
```
C:\Users\varun\anaconda3\Lib\site-packages\sklearn\model_selection\_validation.py:489: FitFailedWarning: 
180 fits failed out of a total of 360.
The score on these train-test partitions for these parameters will be set to nan.
If these failures are not expected, you can try to debug them by setting error_score='raise'.

Below are more details about the failures:
--------------------------------------------------------------------------------
31 fits failed with the following error:
Traceback (most recent call last):
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\model_selection\_validation.py", line 851, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
    ~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\base.py", line 1403, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\pipeline.py", line 649, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
    ~~~~~~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\base.py", line 1393, in wrapper
    estimator._validate_params()
    ~~~~~~~~~~~~~~~~~~~~~~~~~~^^
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\base.py", line 554, in _validate_params
    validate_parameter_constraints(
    ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^
        self._parameter_constraints,
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^
        self.get_params(deep=False),
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^
        caller_name=self.__class__.__name__,
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    )
    ^
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\utils\_param_validation.py", line 98, in validate_parameter_constraints
    raise InvalidParameterError(
    ...<2 lines>...
    )
sklearn.utils._param_validation.InvalidParameterError: The 'criterion' parameter of DecisionTreeRegressor must be a str among {'poisson', 'squared_error', 'absolute_error'}. Got 'friedman_ms
e' instead.

--------------------------------------------------------------------------------
83 fits failed with the following error:
Traceback (most recent call last):
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\model_selection\_validation.py", line 851, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
    ~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\base.py", line 1403, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\pipeline.py", line 649, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
    ~~~~~~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\base.py", line 1393, in wrapper
    estimator._validate_params()
    ~~~~~~~~~~~~~~~~~~~~~~~~~~^^
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\base.py", line 554, in _validate_params
    validate_parameter_constraints(
    ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^
        self._parameter_constraints,
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^
        self.get_params(deep=False),
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^
        caller_name=self.__class__.__name__,
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    )
    ^
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\utils\_param_validation.py", line 98, in validate_parameter_constraints
    raise InvalidParameterError(
    ...<2 lines>...
    )
sklearn.utils._param_validation.InvalidParameterError: The 'criterion' parameter of DecisionTreeRegressor must be a str among {'absolute_error', 'poisson', 'squared_error'}. Got 'friedman_ms
e' instead.

--------------------------------------------------------------------------------
27 fits failed with the following error:
Traceback (most recent call last):
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\model_selection\_validation.py", line 851, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
    ~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\base.py", line 1403, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\pipeline.py", line 649, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
    ~~~~~~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\base.py", line 1393, in wrapper
    estimator._validate_params()
    ~~~~~~~~~~~~~~~~~~~~~~~~~~^^
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\base.py", line 554, in _validate_params
    validate_parameter_constraints(
    ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^
        self._parameter_constraints,
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^
        self.get_params(deep=False),
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^
        caller_name=self.__class__.__name__,
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    )
    ^
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\utils\_param_validation.py", line 98, in validate_parameter_constraints
    raise InvalidParameterError(
    ...<2 lines>...
    )
sklearn.utils._param_validation.InvalidParameterError: The 'criterion' parameter of DecisionTreeRegressor must be a str among {'squared_error', 'absolute_error', 'poisson'}. Got 'friedman_ms
e' instead.

--------------------------------------------------------------------------------
24 fits failed with the following error:
Traceback (most recent call last):
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\model_selection\_validation.py", line 851, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
    ~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\base.py", line 1403, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\pipeline.py", line 649, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
    ~~~~~~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\base.py", line 1393, in wrapper
    estimator._validate_params()
    ~~~~~~~~~~~~~~~~~~~~~~~~~~^^
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\base.py", line 554, in _validate_params
    validate_parameter_constraints(
    ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^
        self._parameter_constraints,
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^
        self.get_params(deep=False),
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^
        caller_name=self.__class__.__name__,
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    )
    ^
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\utils\_param_validation.py", line 98, in validate_parameter_constraints
    raise InvalidParameterError(
    ...<2 lines>...
    )
sklearn.utils._param_validation.InvalidParameterError: The 'criterion' parameter of DecisionTreeRegressor must be a str among {'squared_error', 'poisson', 'absolute_error'}. Got 'friedman_ms
e' instead.

--------------------------------------------------------------------------------
8 fits failed with the following error:
Traceback (most recent call last):
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\model_selection\_validation.py", line 851, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
    ~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\base.py", line 1403, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\pipeline.py", line 649, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
    ~~~~~~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\base.py", line 1393, in wrapper
    estimator._validate_params()
    ~~~~~~~~~~~~~~~~~~~~~~~~~~^^
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\base.py", line 554, in _validate_params
    validate_parameter_constraints(
    ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^
        self._parameter_constraints,
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^
        self.get_params(deep=False),
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^
        caller_name=self.__class__.__name__,
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    )
    ^
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\utils\_param_validation.py", line 98, in validate_parameter_constraints
    raise InvalidParameterError(
    ...<2 lines>...
    )
sklearn.utils._param_validation.InvalidParameterError: The 'criterion' parameter of DecisionTreeRegressor must be a str among {'absolute_error', 'squared_error', 'poisson'}. Got 'friedman_ms
e' instead.

--------------------------------------------------------------------------------
7 fits failed with the following error:
Traceback (most recent call last):
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\model_selection\_validation.py", line 851, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
    ~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\base.py", line 1403, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\pipeline.py", line 649, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
    ~~~~~~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\base.py", line 1393, in wrapper
    estimator._validate_params()
    ~~~~~~~~~~~~~~~~~~~~~~~~~~^^
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\base.py", line 554, in _validate_params
    validate_parameter_constraints(
    ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^
        self._parameter_constraints,
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^
        self.get_params(deep=False),
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^
        caller_name=self.__class__.__name__,
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    )
    ^
  File "C:\Users\varun\anaconda3\Lib\site-packages\sklearn\utils\_param_validation.py", line 98, in validate_parameter_constraints
    raise InvalidParameterError(
    ...<2 lines>...
    )
sklearn.utils._param_validation.InvalidParameterError: The 'criterion' parameter of DecisionTreeRegressor must be a str among {'poisson', 'absolute_error', 'squared_error'}. Got 'friedman_ms
e' instead.

  warnings.warn(some_fits_failed_message, FitFailedWarning)
C:\Users\varun\anaconda3\Lib\site-packages\sklearn\model_selection\_search.py:1234: UserWarning: One or more of the test scores are non-finite: [0.69971309 0.73067083 0.77392016 0.75439731 0.
76310872 0.79444693
 0.80092055 0.80092055 0.80455963 0.82093426 0.82093426 0.82092495
 0.82748587 0.82748587 0.82747656 0.83309144 0.83309144 0.83308018
 0.73820775 0.74189007 0.77798648 0.7559952  0.76574544 0.79648561
 0.80248206 0.80248206 0.80520202 0.69983745 0.73067083 0.77392016
 0.75439731 0.76310872 0.79444693 0.80092055 0.80092055 0.80455963
        nan        nan        nan        nan        nan        nan
        nan        nan        nan        nan        nan        nan
        nan        nan        nan        nan        nan        nan
        nan        nan        nan        nan        nan        nan
        nan        nan        nan        nan        nan        nan
        nan        nan        nan        nan        nan        nan]
  warnings.warn(
```

Out[56]:



In [57]:
```
print("Best DT Parameters:", dt_grid.best_params_)

print("Best DT CV R2 Score:", dt_grid.best_score_)
```
```
Best DT Parameters: {'model__criterion': 'squared_error', 'model__max_depth': 5, 'model__min_samples_leaf': 4, 'model__min_samples_split': 2}
Best DT CV R2 Score: 0.8330914438751318
```

In [58]:
```
dt_pred = dt_grid.predict(X_test)

print(dt_pred[:10])
```
```
[11801.60660648  5243.04129723 26306.769872   11801.60660648
 34232.77860769  5243.04129723  1847.03152738 13995.15160946
  5243.04129723 11801.60660648]
```

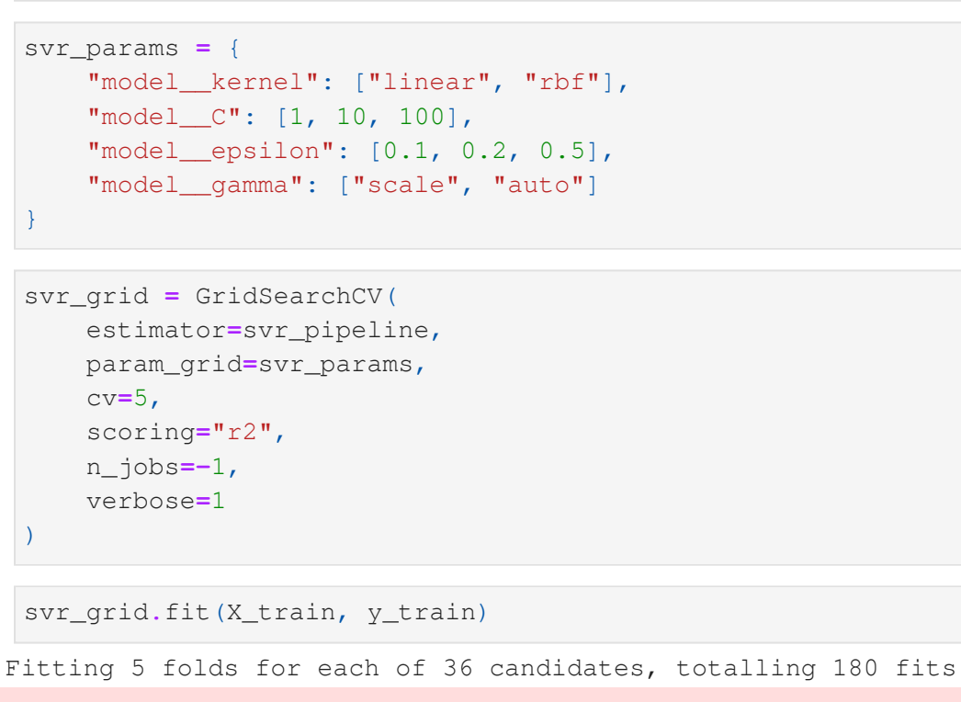
In [59]:
```
svr_pipeline = Pipeline([
    ("preprocessor", preprocessor),
    ("model", SVR())
])
```

In [60]:
```
svr_params = {
    "model__kernel": ["linear", "rbf"],
    "model__C": [1, 10, 100],
    "model__epsilon": [0.1, 0.2, 0.5],
    "model__gamma": ["scale", "auto"]
}
```

In [61]:
```
svr_grid = GridSearchCV(
    estimator=svr_pipeline,
    param_grid=svr_params,
    cv=5,
    scoring="r2",
    n_jobs=-1,
    verbose=1
)
```

In [62]:
```
svr_grid.fit(X_train, y_train)
```
```
Fitting 5 folds for each of 36 candidates, totalling 180 fits
```
```
C:\Users\varun\anaconda3\Lib\site-packages\joblib\externals\loky\process_executor.py:782: UserWarning: A worker stopped while some jobs were given to the executor. This can be caused by a too
short worker timeout or by a memory leak.
  warnings.warn(
```

Out[62]:



In [63]:
```
print("Best SVR Parameters:", svr_grid.best_params_)

print("Best SVR CV R2 Score:", svr_grid.best_score_)
```
```
Best SVR Parameters: {'model__C': 100, 'model__epsilon': 0.1, 'model__gamma': 'scale', 'model__kernel': 'linear'}
Best SVR CV R2 Score: 0.5562170297411093
```

In [64]:
```
svr_pred = svr_grid.predict(X_test)

print(svr_pred[:10])
```
```
[ 9262.59214404  6179.12003344 28427.4086351   9350.25078059
 16617.58221537  5858.22523753  1413.01745849 13430.07801447
  3699.50310253 10268.85709962]
```

In [65]:
```
models = {
    "Random Forest": (rf_grid, rf_pred),
    "Decision Tree": (dt_grid, dt_pred),
    "SVR": (svr_grid, svr_pred)
}

results = []

for name, (grid, predictions) in models.items():

    mae = mean_absolute_error(y_test, predictions)

    rmse = np.sqrt(
        mean_squared_error(y_test, predictions)
    )

    r2 = r2_score(y_test, predictions)

    results.append({
        "Model": name,
        "Best CV R2": grid.best_score_,
        "MAE": mae,
        "RMSE": rmse,
        "Test R2": r2
    })

results_df = pd.DataFrame(results)

print(results_df.sort_values(
    by="Test R2",
    ascending=False
))
```
```
           Model  Best CV R2          MAE         RMSE   Test R2
0  Random Forest    0.838616  2433.385883  4419.815913  0.874171
1  Decision Tree    0.833091  2652.025895  4521.367620  0.868322
2            SVR    0.556217  3627.788029  7716.355228  0.616473
```

In [66]:
```
best_model_name = results_df.sort_values(
    by="Test R2",
    ascending=False
).iloc[0]["Model"]

print("Best Model:", best_model_name)
```

```
Best Model: Random Forest
```

```
In [ ]:
```